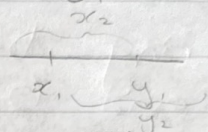


Analysis Tutorial 1

1. Let $A, B \in \mathcal{J}$ with $A = (x_1, y_1]$ and $B = (x_2, y_2]$

Now if $y_1 \leq x_2$ or $y_2 \leq x_1$, then $A \cap B = \emptyset \in \mathcal{J}^+$



So assume $x_2 < y_1$ and $x_1 < y_2$. Then we have the following cases:

- ① $x_2 > x_1$ and $y_2 < y_1$.

Then $A \setminus B = (x_1, x_2] \cup (y_2, y_1] \in \mathcal{J}^+$ as a disjoint union

- ② $x_2 \leq x_1$ and $y_2 < y_1$.

Then $A \setminus B = (y_2, y_1] \in \mathcal{J}^+$

- ③ $x_2 > x_1$ and $y_2 \geq y_1$.

Then $A \setminus B = (x_1, x_2] \in \mathcal{J}^+$

- ④ $x_2 \leq x_1$ and $y_2 \geq y_1$.

Then $A \setminus B = \emptyset \in \mathcal{J}^+$

Thus $A \setminus B \in \mathcal{J}^+$

2. (a) Let A be a ring. Thus $\emptyset \in A$.

Let $A, B \in A$. Since $A \setminus B \in A$, it follows that $A \setminus B \in A^+$. Also, since $A \setminus B \in A$ and $B \setminus A \in A$, we get that $(A \setminus B) \cup (B \setminus A) \in A$ since A is a ring. But then $A \setminus ((A \setminus B) \cup (B \setminus A)) = A \cap B \in A$ since A is a ring. Thus A is a semiring

for $\mathcal{R} = \mathbb{R}$

- (b) Consider $\mathcal{J} := \{(x, y] : x, y \in \mathbb{R}\}$. If $x > y$, then $\emptyset \in \mathcal{J}$. We have shown that $A, B \in \mathcal{J} \Rightarrow A \setminus B \in \mathcal{J}^+$. We must show $A, B \in \mathcal{J} \Rightarrow A \cap B \in \mathcal{J}$

Let $A, B \in \mathcal{I}$ with $A = (x_1, y_1]$ and $B = (x_2, y_2]$. Then $A \cap B = (\max(x_1, x_2), \min(y_1, y_2)] \in \mathcal{I}$ with $A \cap B = \emptyset$ if $\max(x_1, x_2) \geq \min(y_1, y_2)$.

So $\mathcal{I}$ is a semiring, but not a ring because, for example, $(2, 3], (4, 5] \in \mathcal{I}$ and $(2, 3] \cup (4, 5] \notin \mathcal{I}$.

3 (a) Let $\{A_\beta\}_{\beta \in B}$ be a family of rings.

Since $\emptyset \in A_\beta$ for all $\beta \in B$, $\emptyset \in \bigcap_{\beta \in B} A_\beta$.

Now let $A, B \in \bigcap_{\beta \in B} A_\beta$. For each $\beta \in B$, $A, B \in A_\beta$ and since A_β is a ring,

$A \cap B \in A_\beta$ and $A \cup B \in A_\beta$.

So $A \cap B \in \bigcap_{\beta \in B} A_\beta$ and $A \cup B \in \bigcap_{\beta \in B} A_\beta$. So $\bigcap_{\beta \in B} A_\beta$ is a ring.

(b) Consider $A = \{\emptyset, \{1\}, \{1, 2, 3\}, \{2, 3\}\}$

$B = \{\emptyset, \{1\}, \{1, 2, 3\}, \{2\}, \{3\}\}$. Both are semirings, but $A \cap B = \{\emptyset, \{1\}, \{1, 2, 3\}\}$ which is not, because $\{1, 2, 3\} \setminus \{1\} = \{2, 3\}$ cannot be written as a disjoint union.